

LimitIQ

Dynamic Credit Line Management & Exposure Optimization

Executive decision brief

LimitIQ evaluates current limit, +10%, +20% and +30% candidates, then selects the highest eligible simulated risk-adjusted contribution under loss, exposure, payment-history, overextension and profitability controls. Deteriorating accounts are frozen or referred rather than automatically decreased.

Evidence class	Result
Observed source	30,000 Taiwan credit-card accounts; April–September 2005 behavior
Model estimate	Untouched-test ROC-AUC 0.781; Brier 0.133
Simulation	194 increases; INR 454,414 contribution

Portfolio simulation

Accounts	1,200
Increase 10% / 20% / 30%	157 / 37 / 0
Current / proposed limits	INR 478,946,605 / INR 488,028,510
Current / proposed EAD	INR 428,861,224 / INR 436,129,017
Current / proposed expected loss	INR 53,247,037 / INR 53,722,928
Simulated incremental contribution	INR 454,414
Simulated risk-adjusted return	6.25%
Early-warning / manual review	802 / 460

Model governance

Champion: Histogram gradient boosting. The model type and threshold (0.174) were selected using validation data before one untouched-test evaluation. Sex, education, marital status, age, customer ID and target are excluded from decisioning.

Boundaries and committee actions

All line response, drawdown and financial effects are deterministic simulations. PD is not causal. Source TWD monetary values are converted to INR at the documented fixed rate for presentation; this is not Indian borrower evidence. A production build needs current local-law review, verified ability-to-pay inputs, outcome monitoring, model validation, controlled overrides, change approval and rollback. The displayed management ECL is not an IFRS 9 allowance or regulatory-capital calculation.

LimitIQ is an educational portfolio demonstration using public and synthetic data. It is not a production credit-decision system and must not be used to make real lending decisions.